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Measurement of Flow & Viscosity

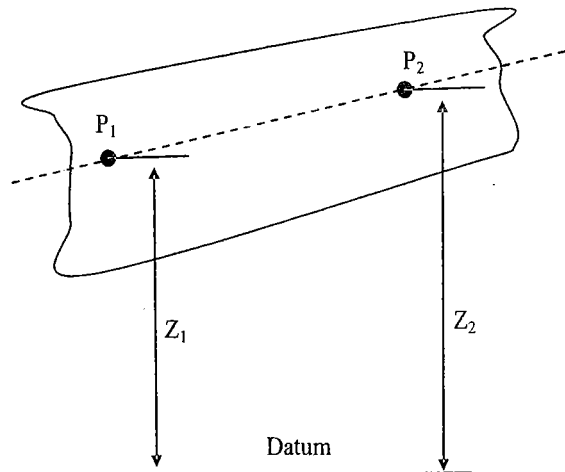
Theory of Variable Head Meters

- The head meter in the form of venturimeter, orifice and flow nozzle is by far the most common flow meter for closed conduits (i.e., pipes). When the fluid flows through these obstruction meters, the fluid accelerates and a reduction in pressure occurs. The difference in pressure before and after the obstruction is measured by means of a differential pressure sensor and is related to the flow rate.
- The equations convenient for practical use and providing the basis of operation of these meters can be derived by following the under given three steps:
 - Write the general energy equation and then modify it for the assumed conditions.
 - Solve the modified equation for down stream velocity.
 - Express the flow rate by the product of velocity and area, i.e.,

$$Q = AV$$
- The equations thus developed will apply equally to any variable head meter. However the individual factors and coefficients differ.
- The energy at any section comprises the internal energy (u), flow work (pv or p/ρ), kinetic energy ($v^2/2$) and gravitational potential energy (gz). Further during the fluid flow between the two chosen sections, heat (q) and mechanical or shaft work (w_s) may also cross the boundaries of the system. Since energy can neither be created nor destroyed, in symbolic form we can write:

$$m_1 \left(u_1 + \frac{P_1}{\rho_1} + \frac{V_1^2}{2} + gz_1 + q \right) = m_2 \left(u_2 + \frac{P_2}{\rho_2} + gz_2 + w_s \right)$$

- Each term in the brackets is expressed per unit mass.
- The symbols used in equation have the following meanings and units:
 - m = fluid mass flow in certain time,
 - u = internal energy,
 - p = static pressure
 - v = specific volume,
 - V = average flow velocity,
 - Z = elevation of centre of fluid above datum,
 - g = local acceleration due to a gravity
 - q = heat entering the system
 - W_s = work output from the system



Flow through a converging section

- Subscript 1 refers to the pipe and the fluid at the upstream pressure connection, subscript 2 refers to the pipe and the fluid at the section of minimum area as indicated in Fig., In actual use, it may not be possible to measure area, pressure, etc., at the smallest section of the stream. In that case subscript 2 refers to the smallest measurable area and to the conditions of fluid at the actual pressure connections. The corrections are then applied



by using a factor called coefficient of discharge.

- We shall make the following assumptions for ease of analysis and for getting simplified results:

- The fluid flow process is adiabatic, i.e., there is no transfer of heat across the system boundaries ($q = 0$)
- no mechanical work is done on or by the fluid ($w_s = 0$)
- the process is frictionless, i.e., $u_1 = u_2$
- the measuring device is horizontal, i.e., the points 1 and 2 are at the same horizontal level ($z_1 = z_2$) and gravity force can be neglected
- there is steady flow of fluid, i.e. the mass flow rate at section 1 is equal to that at section 2, ($m_1 = m_2$)

With these assumptions, the steady flow energy equation can be written as:

$$\frac{P_1}{\rho_1} + \frac{V_1^2}{2} = \frac{P_2}{\rho_2} + \frac{V_2^2}{2} \dots\dots\dots(1)$$

(or)

$$\frac{V_2^2 - V_1^2}{2} = \frac{P_1}{\rho_1} - \frac{P_2}{\rho_2} \dots\dots\dots(2)$$

- The above equation forms the basis of operation of variable head flow measuring devices and is valid both for liquids and ideal gases.
- Further, replacing the density ρ by specific weight by using the relation $w = \rho g$, we get

$$\frac{P_1}{w_1} - \frac{P_2}{w_2} = \frac{V_2^2 - V_1^2}{2g} \dots\dots\dots(3)$$

- This is well-known Bernoulli's equation, each term has units of length m.

Variable head-meters for incompressible fluids

The continuity relation for the flow situation is

$$m = \rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

Where $\rho_1 = \rho_2 = \rho$ for incompressible fluid flow from section 1 to section 2. This yields

$$V_1 = (A_2/A_1)V_2 \dots\dots\dots(4)$$

Solution of equations (1) and (4) gives out flow velocity

$$V_2 = \frac{1}{\sqrt{1 - (A_2/A_1)^2}} \sqrt{\frac{2(P_1 - P_2)}{\rho}} \dots\dots\dots(5)$$

- The downstream fluid velocity V_2 as given by equation shown above has been derived without considering any losses and so it is the ideal or hypothetical velocity at the minimum section. The actual velocity can be obtained by multiplying the theoretical velocity by a factor C_v called the coefficient of velocity. The coefficient of velocity is the ratio of actual mean velocity to the ideal mean velocity which would occur without any friction loss.

$$C_v = \frac{\text{actual mean velocity}}{\text{ideal mean velocity}}$$

$$\therefore V_2(\text{actual}) = C_v \frac{1}{\sqrt{1 - (A_2/A_1)^2}} \sqrt{\frac{2(P_1 - P_2)}{\rho}}$$

- For obtaining the volume flow rate, we apply the continuity equation.

Discharge = area $\times$ velocity

$$\therefore Q = A_2 C_v \frac{1}{\sqrt{1 - (A_2/A_1)^2}} \sqrt{\frac{2(P_1 - P_2)}{\rho}}$$

- During flow through an orifice meter, the fluid jet on leaving the orifice contracts to minimum area at a section called vena-contracta. Area of fluid jet at vena contracta is less than area of the orifice and the two are related by the equation:



- Area of jet at vena contracta = $C_c \times$ orifice area
- Where C_c is the co-efficient of contraction. Thus if the orifice area is A_2 , then the area at minimum section which controls the flow rate and where the path of particles becomes parallel again would be $C_c A_2$.

$\therefore$ Actual discharge

$$Q = (C_c A_2) C_v \frac{1}{\sqrt{1 - (A_2 / A_1)^2}} \sqrt{\frac{2(p_1 - p_2)}{\rho}}$$

- For venturi meter and flow nozzle, there is almost no formation of vena contracta and the coefficient of contraction can be taken as unity.
- Combining C_v and C_c into a single factor C_d called the coefficient of discharge, the volumetric flow rate through the meter can be written as

$$Q = C_d \frac{A_2}{\sqrt{1 - (A_2 / A_1)^2}} \sqrt{\frac{2(p_1 - p_2)}{\rho}} \dots\dots\dots(6)$$

- The discharge coefficient C_d is not constant; it depends primarily on the flow Reynolds number and the channel geometry.
- For given meter, A_1 and A_2 have definite values and so the quantity $\sqrt{1 - (A_2 / A_1)^2}$
- Is constant for the meter. This parameter allows for the additional kinetic head of the approaching fluid and is called velocity of approach factor ; its value is always greater than unity. For upstream velocity V_1 equal to zero, the velocity of approach factor $M=1.0$.

$$Q = C_d M A_2 \sqrt{\frac{2(p_1 - p_2)}{\rho}}$$

- For venture meter, the factors C_d and M are applied separately but for nozzles and orifice plates these factors are combined into a single

factor known as the flow coefficient and designated as K .

$$\therefore Q(\text{actual}) = C_d M A_2 \sqrt{\frac{2(p_1 - p_2)}{\rho}}$$

For venturi meter

$$= K A_2 \sqrt{\frac{2(p_1 - p_2)}{\rho}}$$

For nozzles and orifice plates

- The area A_2 is usually determined from measurements obtained at room temperature but in use this area is influenced by the temperature of the fluid flowing through the meter. The actual discharge is essentially worked out from the relation

$$Q(\text{actual}) = C_d M A_2 E \sqrt{\frac{2(p_1 - p_2)}{\rho}} \text{ for}$$

venturi meter.

$$= K A_2 E \sqrt{\frac{2(p_1 - p_2)}{\rho}}$$

for nozzles and orifice plates.

- Where E is thermal expansion correction factor. If the density is replaced by specific weight w , equation can be written as:

$$Q(\text{actual}) = C_d M A_2 E \sqrt{\frac{2g}{w} (p_1 - p_2)}$$

for venturi meter

$$= M A_2 E \sqrt{\frac{2g}{w} (p_1 - p_2)}$$

for nozzles and orifice plate.

- The pressure differential $\frac{(p_1 - p_2)}{w}$ called pressure head or piezometer head, is measured by a differential manometer, pressure gauge or any other pressure measuring device. The gauge reading is same for a given discharge irrespective of the inclination of the pipe line. This can be shown by considering the measuring device laid in horizontal, inclined or vertical position.

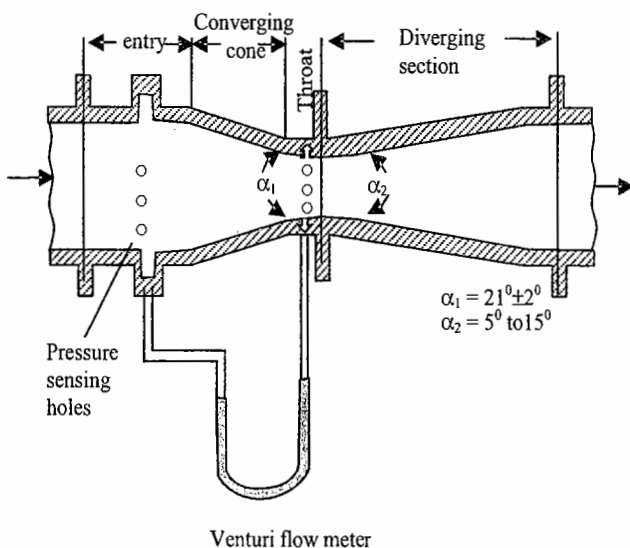
Venturi flow meter

- This simple and reliable device finds an extensive use for water flow measurement, particularly in large size pipes and for large flow rates.

The important constructional features of a venturi meter are shown in the meter consists of

Cylindrical entrance section:

- This section has the size of the pipe to which it is attached. For satisfactory operation, the venturi meter should be preceded by a straight pipe of not less than 5 to 10 pipe diameters and free from fittings, misalignment and other sources of large scale turbulence. If these conditions cannot be met, straightening vanes should be placed upstream from the meter for reduction of rotational motion in flow.



Venturi flow meter

Converging conical section

- The converging takes place at an angle of $21^\circ \pm 2^\circ$. The velocity of fluid increases as it passes through the converging section and correspondingly the static pressure falls.

Throat:

- This is a cylindrical section of minimum area. At this section the velocity is maximum and the pressure is minimum. The throat diameter is usually between $\frac{1}{2}$ to $\frac{1}{4}$ of the inlet diameter. The length of the throat equals its diameter

Diverging section:

- in which there is a change of stream area back to the entrance area. The recovery of kinetic energy by its conversion to pressure is nearly complete and so the overall pressure loss is small. To accomplish a maximum recovery to kinetic energy, the diffuser section is made with an included angle of 5° to 7° . The angle has to be kept less so that the flowing fluid has least tendency to separate out from the boundary of the section. However, with small angles, the length and hence the cost of the meter would increase. So where pressure recovery is not of much importance, the angle of the diverging cone may be kept as high as 14° .
- The pressure recovery is not of much importance, the angle of the diverging cone may be kept as high as 14° .
- The pressure taps are made at the throat and at entrance where the venturi has a diameter equal to that of pipe. The pressure taps may be made from a single hole at piezometer rings thereby giving average pressure values at the two sections.
- The small sized venturi meters, suitable for pipe lines less than 5cm in diameter, are usually made of brass or bronze. The inside surface is smoothly finished to reduce friction. Large venturies are usually made of cast iron; the throat is however lined with brass or bronze and machined to a smooth finish. Very large venturies upto 6m pipe



diameter have been made of smooth surface concrete, the throat only being of machined bronze.

Advantages and limitations:

- High pressure recovery is attainable, i.e., loss of head due to its installation in the pipe lines is small.
- Because of smooth surface, the meter is not much affected by wear and abrasion.
- Well established characteristics; years of applicational experience.
- Less likelihood of becoming clogged with sediments.
- Due to low value of losses, the co-efficient of discharge is high and approaches unity under favourable conditions
- Long laying length; space requirements are more
- Quite expensive in installation and replacement
- Difficult to construct particular so as to provide reproducibility.

Venturi meters are ideally suited for measurement of :

- Large flow of water
- Process fluids, i.e., fluids used in industrial processes
- Industrial wastes
- Gases
- Suspended particles, slurries and dirty liquids
- Venturimeter are not standardized yet to an extent that permits discharge coefficient obtained conditions, the discharge coefficient for a venturimeter is found to be function of Reynolds number and meter size. These meters are not generally useful below 7.5cm pipe diameter.

Example 1:

A horizontal venturimeter with 15cm inlet and 7.5cm throat is used for measuring the flow of cooling water in a manufacturing process. The differential pressure between the inlet and throat is 17.5cm when measured using a mercury U-tube manometer. Make calculation for the water flow rate if the discharge coefficient for the venturimeter is 0.97. (Density of water = 1000kg/m^3 and specific gravity of mercury = 13.6).

Ans: = $0.02915\text{ m}^3/\text{sec}$

Sol: given

$$D_1 = 15\text{ cm}$$

$$A_1 = 0.017\text{ m}^2$$

$$D_2 = 7.5\text{ cm}$$

$$A_2 = 0.0044\text{ m}^2$$

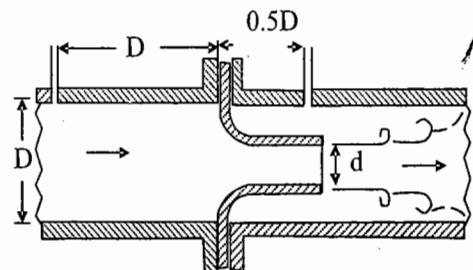
$$C_d = 0.97$$

We know

$$\begin{aligned} Q &= C_d \cdot M \cdot E \sqrt{2g(sg-1) \times h_m} \\ &= 0.97 \times \frac{0.017 \times 0.0044}{\sqrt{(0.017)^2 - (0.0044)^2}} \\ &\quad \times \sqrt{2 \times 9.8 \times 12.3 \times 17.5 \times 10^{-12}} \\ &= 0.02915\text{ m}^3/\text{sec} \end{aligned}$$

Flow Nozzle

- Nozzles are used in engineering practice for the creation of jets



Flow nozzle

and streams for all purposes as well as for fluid metering. When placed in or at the end of a pipe line as metering devices, they are called flow nozzles.



- The flow nozzles comprises a smooth, gradual contraction to throat followed by a free, uncontrolled expansion back to the original pipe flow area. Because the device contains no provision an orderly transformation of velocity pressure into static pressure, the nozzle has a pressure loss of from 80 to 90% of the differential pressure obtained

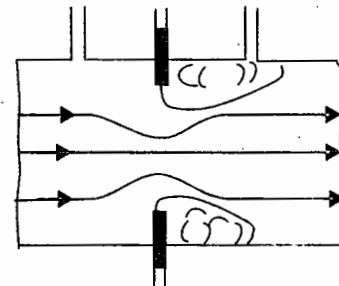
Advantages and Limitations:

- Cheaper than a standard venturi meter and can be installed in an existing main without great difficulty.
- Increased coefficient of discharge when compared with orifice.
- Physical length is less compared to venturi meter.
- Widely accepted for high-pressure/temperature steam flow.
- Good for fluid containing solids that settle.
- Discharge of liquids into the atmosphere can be metered. The down stream pressure p_2 in that case is atmospheric pressure and we need
- to measure the difference between upstream pressure p_1 and atmospheric pressure.
- Pressure recovery is poor and so cannot be used where available pressure head is small or where pressure recovery is a must. The pressure loss is, however, less than that of an orifice plate.
- Compared to orifice meter, it is expensive and also difficult to install.
- Limited to moderate pipe size ; not available above 120cm.

Orifice flow Meter

- The orifice meter consists of a thin, circular metal plate with a hole in it. The plate is held in the pipe line between two flanges called orifice flanges. The flow characteristics of the

orifice differ from those of a nozzle in that minimum section of the stream tube occurs not within the orifice but downstream from the orifice edge. This minimum area is called vena contracta and it is at this area where minimum pressure is obtained. It is not feasible to locate the down stream pressure connection at the vena contracta because the location of vena-contracta depends on Reynolds number, area



Orifice plate

- Ratio between orifice and pipe, the roughness of the pipe and the compressibility of the fluid. It is also sensitive to the upstream velocity profiles and to the sharpness of the upstream edge of the orifice plate. Accordingly downstream pressure tapping is taken at a fixed proportion of the pipe diameter and correction for vena -contracta is made.

Advantages and limitations

- Low initial ease of installation and replacement.
- Characteristics are well known/ and predictable ; years of applicable experience.
- Requires less space as compared with venturi meter
- Can be used in wide range of pipe size (1.25cm to 150cm)
- Pressure recovery is poor. The overall pressure loss varies from 40 to 90% of the differential pressure.
- Necessity of providing straightening vanes upstream.



- Coefficient of discharge has a low value.
- Success of providing straightening vanes upstream.
- Tends to clog; not suitable for slurries or entrained particles.
- More likely to be damaged by pressure transient because of its lower physical strength
- The wide application of the variable head meters is due to following reasons:-
- Possibility to meter all types of liquids
- No moving parts immersed in the liquid being metered; mechanical wear and tear is practically absent.
- Possibility to build the meter to suit any shape of the conduct which need not be horizontal or vertical
- No limit to the maximum flow that can be metered.
- Possibility of indicating, diagrammatic record and integration of the flow rate.
- Possibility to transmit the measured signal over long distance.

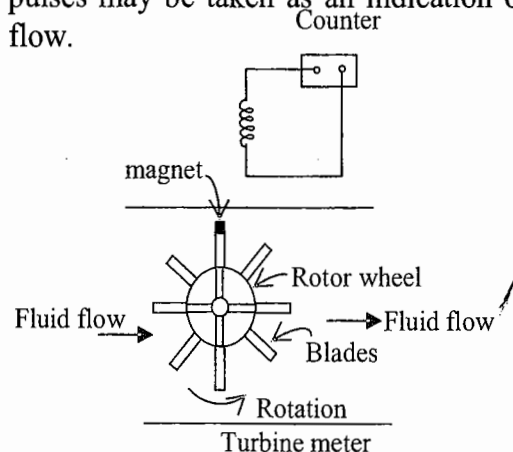
Pitot tubes, anemometers current meters:

- Flow velocity constitutes an important parameter in kinematics and dynamics of fluid flow and, therefore, its measurements is quite essential. Velocity measurement may be made with a view to :
- Determine the volumetric flow rate.
- Forecast the weather from wind velocity measurement.
- Locate the separation points from a knowledge of velocity distribution.
- Flow velocity measurements may be classified as:
 - Instruments like the pitot-tube and hot wire anemometers which measure the local velocity at a point in the channel through which the fluid is flowing

- Instruments like the cup and vane anemometers, the current and turbine meters which measure the average velocity of flow.

Turbine flow Meter

- It consists of a freely rotating wheel (rotor or propeller) with multiple blades. The rotor is supported by ball or sleeve bearings, and is located centrally in the pipe along which the flow occurs. The flowing fluid impinging on the turbine blades imparts a force on the blade surfaces and sets the rotor in motion with an angular speed proportional to the fluid velocity. The rotor speed is measured with a mechanical counter or with an electro-magnetic pick up and associated counter. The electro-magnetic pick up could be a small permanent magnet mounted at the tip of one of the rotor blades with a coil being placed just outside the tube. As the magnet moves past the coil, an e.m.f. is induced. The voltage pulse is picked up for each rotation of the wheel. Faster the fluid flow, the greater the count per second. Each pulse represents a definite flow quantity, and the total number of pulses may be taken as an indication of total flow.



$$Q = \frac{f}{K}$$

- Where f is the pulse frequency, Q is the volume flow rate and K is the flow coefficient

that depends on the flow rate and the viscosity. By minimizing the bearing friction and other losses, the device can be designed to give a linear output.

Advantages and limitations:

- Good accuracy, excellent repeatability and wide range commencing from 0.5 lpm to 15,000 lpm for liquids and 3 lpm to 500,000 lpm for air
- Low pressure drop, good temperature and pressure ratings
- Easy to install and maintain
- Adaptability to flow totalizing and to digital blending systems
- Good transient response
- Relatively high cost and limited use for slurry applications
- Requires in-line mounting and a straight run of pipe (15 diameter) ahead of the meter.

Rota Meter

The Rota meter consists of a tapered metering glass tube, inside which is located the rotor or active element (float) of the rotor. This tapering tube is provided with suitable inlet and outlet connections. The float or bob material has specific gravity higher than that of the fluid to be metered.

- With increase in the flow rate, the float rises in the tube and there occurs an increase in the annular area between the float and the tube. The float adjusts its position in relation to discharge through the passage i.e., the float rises higher or lower depending on the flow rate. The discharge equation for flow through a Rota meter is given by

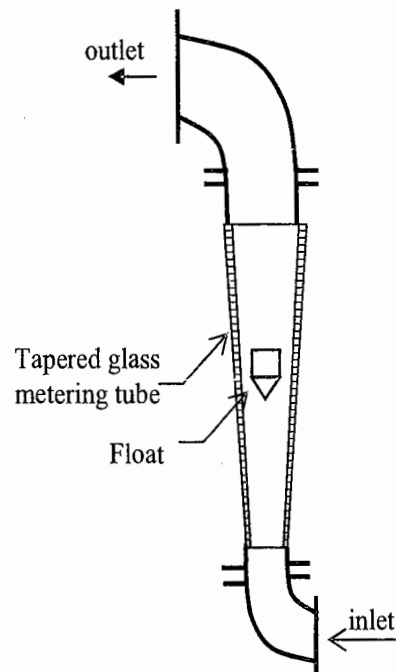
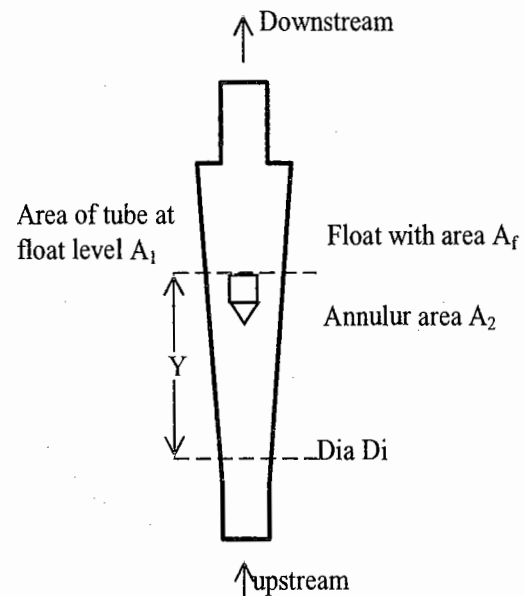
$$Q = C_d A [2gV_f (\rho_f - \rho) / A_f \rho]^{\frac{1}{2}}$$

- Where Q is the volume flow rate, C_d is the discharge coefficient, V_f is the volume of

float, ρ_f is the density of float material, ρ is the density of fluid flowing, A_f is the cross-sectional area of the float and A is the annular area between the float and tube.

- The relationship between discharge and the float position can be worked out by carrying out the following analysis.

Variable-area flow meters





Weight of float = $\rho_f V_f g$

Buoyant force on the float = $\rho V_f g$

- Drag force on the float = $A_f (p_1 - p_u)$ where p_1 and p_u are the pressure on the lower (upstream) and upper (downstream) face of the Rota meter.
- For static equilibrium of float at any position, there is balance between the drag force, buoyant force and weight of float.

That is $A_f (p_1 - p_u) + \rho V_f g = \rho_f V_f g$

That gives $(p_1 - p_u) = \frac{V_f}{A_f} g(\rho_f - \rho)$

- The two surfaces of the float form a kind of constriction (obstruction) and accordingly

$$Q_{act} = C_d \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \sqrt{2g p_h}$$

Where p_h = Piezometric head

$$= \sqrt{\frac{p_1 - p_u}{\rho g}} = \sqrt{\frac{V_f (\rho_f - \rho)}{A_f \rho}}$$

$A_1 = A_t$ = area of tube at float level

$A_2 = A_t - A_f$ = minimum annular area between the tube and float

C_d = coefficient of discharge

Therefore

$$Q_{act} = C_d \frac{A_t (A_t - A_f)}{\sqrt{A_t^2 - (A_t - A_f)^2}} \sqrt{2g} \times \sqrt{\frac{V_f (\rho_f - \rho)}{A_f \rho}}$$

- If the discharge coefficient is assumed to remain constant with the position of float and further if

$(A_t - A_f)/A_t \ll 1$, then $Q_{act} = K(A_t - A_f)$

Where

$$K = 2g \times \sqrt{\frac{V_f}{A_f} \left(\frac{\rho_f - \rho}{\rho} \right)}$$

is the Rota meter constant.

Let θ be the taper of the Rota meter tube, then

$$A_t = \frac{\pi}{4} (D_i + y \tan \theta)^2 = \frac{\pi}{4} D_i^2 + \frac{\pi}{2} y D_i \tan \theta$$

- Where D_i is the diameter at inlet and y is the float position with respect to inlet

$$\begin{aligned} Q_{act} &= K \left[\frac{\pi}{4} D_i^2 + \frac{\pi}{2} y D_i \tan \theta - A_f \right] \\ &= K \frac{\pi}{4} y D_i \tan \theta + K \left[\frac{\pi}{4} D_i^2 - A_f \right] \\ &= K_1 y + K_2 \dots \dots \dots (13) \end{aligned}$$

- The constants K_1 and K_2 depend upon the shape of Rota meter and the float. Equation(13) shows that there exists a liner relationship between the flow rate and displacement of float in case of a Rota meter..
- The glass tube is often made of high strength borosilicate glass and the flow rate scale is engraved on the tube corresponding to a particular float material. However, where greater strength is required, the metallic tubes are used and the position of float is detected magnetically. Further, for remote indication the position of float can be monitored electrically by using a suitable displacement transducer.
- The Rotameter is equally suitable for measuring both gas and liquid flows. When metering gas flows a small sphere is used in a narrow tube and no guidance needs to be provided to its motion. While metering liquid fluids, spherical slots are cut no a part of the float and these slots cause it to rotate slowly about the axis of tube and keep it central. This spinning helps also to prevent accumulation of any sediment on the top and sides of the float. Alternatively, stability of the bob is

ensured by employing a guide along which the float would slide.

- The float or bob material has a specific gravity higher than that of the fluid to be metered. The density difference ($\rho_f - \rho$) required for metering a particular liquid or gas can be obtained by selecting different materials for the float. An examination of the discharge equation would reveal that when $\rho_f \gg \rho$, the volume flow rate becomes independent of the density ρ of the flowing fluid.
- Rota meters are widely used for metering purge flows, pump-seal fluids and coolants and lubricants for operating machinery. In these applications, flows are relatively small and accuracy requirement are not rigid.

Advantages and limitations

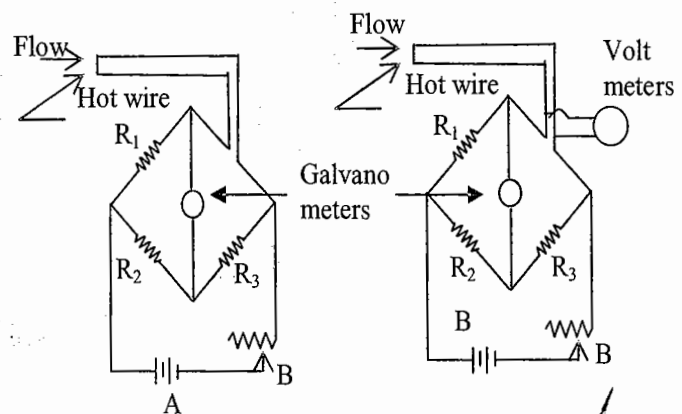
- Simplicity of operation, ease of reading and installation
- Relatively low cost
- Handles wide variety of corrosive fluids
- Easily equipped with data transmission, indication and recording devices
- Possibility of convenient and visible flow comparisons by mounting several Rota meters side by side
- Glass tube subject to breakage
- Limited to small pipe sizes and capacities
- Less accurate compared to venturi and orifice meters
- Must be mounted vertically
- Subject to oscillations in pulsating flows

Hot wire anemometer

- Hot-wire anemometry has been considered as satisfactory approach to the measurement of the mean and fluctuating flow components in a flow field. The sensor is a 5 micron

diameter platinum-tungsten wire welded between the two prongs of the probe and heated electrically as part of a wheat-stone bridge circuit,. When the probe is introduced into the flowing fluid, it tends to be cooled by the instantaneous velocity and consequently there is a tendency for the electrical resistance to diminish. The rate of cooling of wire depends upon the

- Dimensions and physically of wire,
- Difference of the temperature between the wire and the fluid,
- Physically properties of the fluid, and
- Stream velocity under measurement. For a simple hot wire anemometer the first three conditions are effectively constant and the instrument. For a simple hot wire anemometer the first three conditions are effectively constant and the instrument response is then a direct measurement of the velocity.



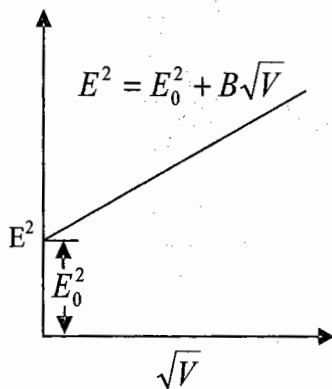
Hot wire bridge circuits

- Depending on the associated electronic equipment, hot wire may be operated in two ways. **Constant-current mode** in this mode the heating current, i.e., voltage across the bridge is kept constant. Initially the circuit is adjusted such that the galvanometer reads zero when the heated wire lies in stationary air. When the air flows, the hot wire cools; the

resistance changes and the galvanometer deflects. The galvanometer deflections are amplified, measured and correlated with air velocity by previous calibration.

Constant-temperature mode:

- In this mode the operating resistance of the wire and hence its temperature is maintained constant. In the event of the tendency of the hot-wire to cool by the flowing fluid, the external bridge voltage is applied to the wire to maintain a sensibly constant temperature. The bridge voltage is varied so as to bring the galvanometer needle to zero ; the reading on the voltmeter is recorded and correlated with air velocity.
- The response of a hot wire placed normal to the flow velocity V is known to be described by :



Response of hot wire

$$E^2 = E_0^2 + BV^n$$

- Where E is the instantaneous value of the bridge voltage, E_0 and B are the constants depending on the cold resistance of the wire and on the properties of the fluid medium. The values of these constants for a given wire are found by calibrating the wire in a uniform flow field against a standard pitot-static probe. Generally with exponent $n = 0.5$, the response of the hot wire in the form of a plot between E^2 and V^n is a straight line.

Electro Magnetic Flow Meter

- The operation of electromagnetic flow-meters is based on Faraday's law of electromagnetic induction, which states that an emf of e volts is induced in a conductor of length l metres when it moves with a transverse velocity of v m/s across a magnetic field of strength B wb/m² and the induced emf in the conductor is given by the expression $e = B/v$ volts
- Since the voltage induced depends on the rate at which the conductor moves through the magnetic field, the magnitude of voltage induced can be used as an indication of the flow-rate of liquid.
- The flowing fluid through the pipe should be conductive and should have a conductivity at least about 10^{-3} mho per meter.
- As the conductive fluid flows through the insulated tube through the magnetic field, a voltage is induced across the electrodes.
- The voltage induced across the electrodes varies directly in proportion to
 - the magnetic field strength, B
 - the distance between the electrodes, d
 - the average velocity of the conductive fluid flowing through the non-conductive tube.
- If the magnetic field strength B and the distance between the electrodes d are kept constant the voltage induced will be directly proportional to the flow-rate of the flowing fluid.

Advantages:

- The main advantages of the electromagnetic flow-meter are that it does not obstruct the flow in any way that may cause pressure drops.
- It has no moving part that may cause friction
- It is insensitive to viscosity, density, and flow disturbances (as long as the velocity profile is symmetrical).

- It has a wide linear range of 10 : 1 with good accuracy and reliability.
- It is simple and rugged in construction.
- Its response is fast and is independent of the properties of the fluid except for the electrical conductivity.
- The only condition is that the pipe should always be full.
- The electromagnetic flow-meter is ideal for bidirectional measurements and is well suited for slurries, and corrosive and solid contaminated liquids.

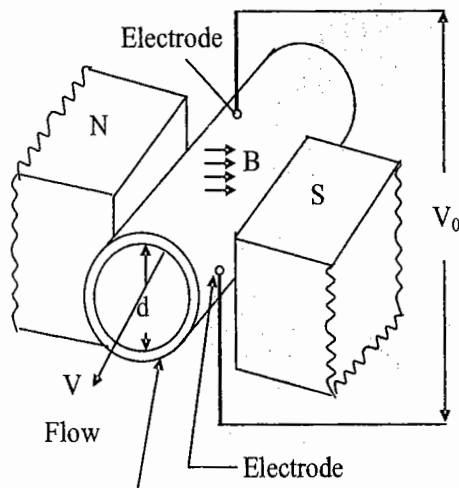


Fig. Electromagnetic flow-meter

Example 2:

Calculate the generated emf (rms value) by the magnetic flow meter if the ordinary tap water flows at the rate of 400 liters per minute through its tube of 8 cm diameter. Assume the maximum flux density of the alternating magnetic field established in the flow meter to be 400 lines per cm^2 (gauss).

Ans: 6.0 mv

Sol: Given, $Q_v = \frac{400}{60} = 6.667 \text{ litre/sec}$

$l = 8 \text{ cm} = 0.08 \text{ m}, B = 400 \text{ gauss}$
 $= 400 \times 10^{-4} \text{ tesla}$

We know,

$$E = \frac{4B}{\pi l} Q_v \times 10^{-3} \text{ V} \approx 4.24 \text{ mv}$$

$$E_{\text{rms}} = \frac{2 \times 4.24}{\sqrt{2}} \text{ mv} = 6.0 \text{ mv}$$

Ultrasonic flow meter

- Basically an ultrasonic or acoustic flow-meter consists of two piezoelectric crystals in the liquid or gas separated by a distance. One of the crystal acts as a transmitter and the other as a receiver.
- These flow meters are of two types depending upon the principle on which they operate.
- The first one operates on the basis of measurement of phase shift between oscillations directed downstream and upstream alternately while the second one operates on the basis of measurement of separation of frequency of oscillations directed downstream and upstream simultaneously.
- The schematic diagram of an ultrasonic flow meter operating on first principle, called the "Doppler effect", is shown in Fig. below

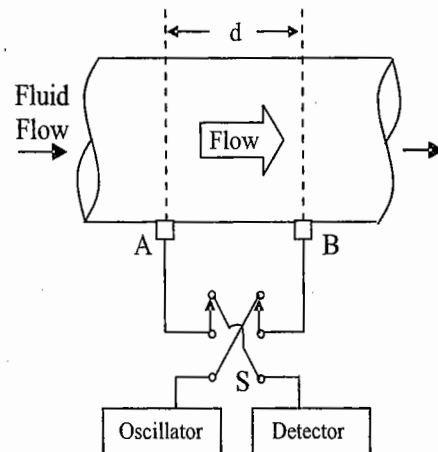


Fig. Schematic Diagram of Ultrasonic Flowmeter Operating

- Two piezo crystals A and B working both as transmitter and receiver of ultrasonic signals

alternately are mounted so that the ultrasonic signals are transmitted between them, as well as through the liquid.

- An oscillator supplies alternately A or B through the switch S when the detector is connected simultaneously to B or A respectively.
- The detector is a phase sensitive device which measures the transit time from upstream to down stream and vice-versa.
- If C is the velocity of sound propagation in medium in m/s, v is the linear velocity of fluid flow in m/s and d is the distance in metres between the two piezo crystals A and B. The transit time in the direction of flow (i.e., from upstream to down stream)

$$\Delta t_1 = \frac{d}{C+v} \dots\dots\dots (i)$$

- The transit time when the ultrasonic signal is traveling in the opposite direction of fluid flow.

$$\Delta t_2 = \frac{d}{C-v} \dots\dots\dots (ii)$$

- Similarly a sinusoidal signal of frequency f Hz traveling in the direction of fluid flow has a phase shift of

$$\Delta \phi_1 = \frac{2\pi f d}{C+v} \text{ radians } \dots\dots(iii)$$

and that traveling against the direction of fluid flow has a phase shift of

$$\Delta \phi_2 = \frac{2\pi f d}{C-v} \text{ radians } \dots\dots(iv)$$

- The difference between two phase shifts is proportional to the flow-rate of the fluid as obvious from the following expression.

$$\begin{aligned} \Delta \phi_2 - \Delta \phi_1 &= \frac{2\pi f d}{C-v} - \frac{2\pi f d}{C+v} \\ &= \frac{4\pi f d v}{C^2 - v^2} = \frac{4\pi f d v}{C^2} = Kv \dots (v) \end{aligned}$$

As the fluid flow velocity $\ll C$, the acoustic velocity.

- The linearity is good – within 2% and linear flow velocity upto about 1mm/s in the low side can be measured easily.
- The dynamic response is limited by the switching frequency. However the uncertainty of the value of C may be a source of an error.
- Flow channel walls should be such as not to permit acoustic transmission. Plastic materials may be used for this purpose.
- The fundamental frequency is determined by the thickness of the piezo crystal. Probes of different frequencies are employed for different groups of fluids, the normal value being in the range of 200 to 5000 kHz.
- The schematic diagram of an ultrasonic flow-meter operating on second principle is illustrated in Fig. Such a flow-meter is called the transit time flow-meter or leading-edge meter.
- In this arrangement two sets of piezo crystals (a transmitter and a receiver as a set) are mounted as shown in the fig. and wave trains are sent at an angle θ with the direction of fluid flow in either way.

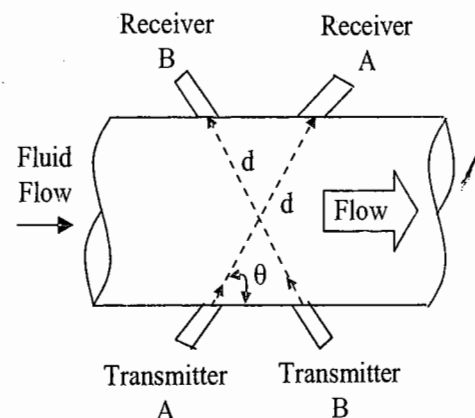


Fig. Schematic Diagram of Transit Time Ultrasonic Flowmeter



- The velocity of the ultrasonic signal from the transmitter A to the receiver A is increased while that from the transmitter B to receiver B is decreased because of fluid flow velocity in the direction or in opposite direction of sound path.
- If C is the sound propagation velocity in medium in m/s, v is the fluid flow velocity in m/s, d is the distance between the transmitter and receiver in meters and θ is the angle between the sound path and the pipe wall, the repetition frequencies in the upstream side and downstream side will be

$$f_a = \frac{C + v \cos \theta}{d} \text{ and } f_b = \frac{C - v \cos \theta}{d}$$

respectively.

The difference in frequency is given by

$$\Delta f = f_a - f_b = \frac{2 v \cos \theta}{d} \dots\dots\dots (vi)$$

- By measuring the difference in the repetition frequency Δf , and knowing the values of θ and d , the velocity of the fluid flow can be determined. Alternatively, the flow velocity can be determined by measuring the time difference between the two pulse trains in either direction as $\Delta t = \frac{1}{\Delta f}$

Advantages:

- Expression shows that output is independent of C , therefore the effects of pressure and temperature are avoided.
- Also the output is linearly proportional to the velocity of fluid flow.

- The measurement is insensitive to viscosity, pressure and temperature variations.
- The other advantages are no obstruction to flow, bidirectional measuring capabilities, good accuracy, fast response, wide frequency range and its versatility in that it can be employed for any pipe size.

Disadvantages:

- However the complexity and relatively high cost limit its use for individual applications.

Example 3:

In an ultrasound flow meter the transducers are separated by a distance $l = 25 \text{ mm}$ with the line joining the transducers making an angle of 60° with the direction of flow. The transit time difference between upstream and downstream measurements is 10 ns with the sound velocity in the medium being 1000 m/s . Assuming that the size of the transducers is very small as compared to the diameter of the pipe, what is the flow velocity?

Ans: 0.4 m/sec

Sol: Given $\Delta t = 10 \times 10^{-9} \text{ sec}$

$$V_s = 1000 \text{ m/sec}$$

$$d = 25 \text{ mm}$$

$$\theta = 60^\circ$$

$$\begin{aligned} \text{So, } V_f &= \frac{\Delta t \times V_s^2}{2d \cos \theta} \\ &= \frac{10 \times 10^{-9} \times (1000)^2}{2 \times 25 \times 10^{-3} \times \cos(60^\circ)} \\ &= 0.4 \text{ m/sec} \end{aligned}$$